

Essa on AWS — Plans and Pricing

Database and application hosting in Mumbai (ap-south-1)

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Three sizing options for running the Essa ERP, POS and document-intake system on AWS, costed line by line at published Mumbai on-demand rates. **Plan B is the recommendation** — it lands at roughly ₹10,600 a month all-in, which is the budget you set, and it has headroom to grow into. Every rate below was read from AWS's own pricing pages on 4 September 2026.

1 How AWS charges — and what you are actually buying

There is no "database plan" to purchase. AWS bills metered components by the hour or the gigabyte, and a monthly figure is the sum of five of them:

Component	What it is	Mumbai rate
Database compute	The RDS instance, billed per hour it exists — running or idle	\$0.021–\$0.167 /hr by class
Database storage	Provisioned gp3 disk, whether or not it is full	\$0.131 per GB-month
Backups	Free up to 100% of provisioned storage; charged beyond it	\$0.095 per GiB-month
Application compute	The EC2 server running FastAPI, the POS and the phone app, plus its EBS disk	\$0.0112–\$0.0448 /hr
Data transfer out	Bytes leaving AWS for the internet. Inbound is free.	100 GB/mo free, then \$0.1093/GB

Two consequences worth internalising. Stopping an RDS instance stops the compute charge but not the storage or backup charge — and RDS restarts a stopped instance automatically after seven days. And a Multi-AZ deployment costs exactly twice a Single-AZ one on both compute and storage, because AWS runs a full standby copy.

SIZING RATIONALE

Essa is roughly 100 employees on a warehouse app with 50 data models, a Flask POS in its own schema, and invoice images arriving through a vision model. That is a read-heavy transactional workload with a busy hour at shop opening and at month-end, not a continuous high-concurrency load. It fits a burstable (T-class) instance comfortably, and Graviton (t4g) is the right family — in Mumbai a `t4g.medium` costs \$0.0224/hr against \$0.0448/hr for the equivalent `t3.medium`, exactly half, for the same 2 vCPU and 4 GiB.

2 Plan A — Pilot **SINGLE SHOP / PROOF OF RUN**

One shop live, a handful of concurrent users, small document volume. Enough to migrate off Supabase and run real transactions, not enough for the whole business.

Line item	Specification	Calculation	USD/mo	₹/mo
Amazon RDS for PostgreSQL	db.t4g.micro, Single-AZ	$\$0.021 \times 730 \text{ hr}$	15.33	1,449
RDS storage	20 GiB gp3	$20 \times \$0.131$	2.62	248
RDS automated backups	within 100% of storage	included	0.00	0
EC2 application server	t4g.small, Linux	$\$0.0112 \times 730 \text{ hr}$	8.18	773
EBS volume	30 GiB gp3	$30 \times \$0.0912$	2.74	259
EBS snapshots	~20 GiB retained	$20 \times \$0.05$	1.00	94
S3 (invoice scans)	50 GB Standard	$50 \times \$0.025$	1.25	118
S3 requests	light volume	estimate	0.30	28
Public IPv4 address	1 address	$\$0.005 \times 730 \text{ hr}$	3.65	345
Data transfer out	~50 GB	under 100 GB free tier	0.00	0
Total, on-demand, before tax			\$35.06	₹3,313
Including 18% GST			—	₹3,910

WHERE IT RUNS OUT

1 GiB of database memory caps you at roughly 112 connections and leaves little room for the query planner. Month-end reporting across a year of GRNs will feel it. Treat Plan A as a pilot with a known upgrade path — changing instance class later is a reboot, not a migration.

3 Plan B — Production **RECOMMENDED**

The whole business live: warehouse, POS, phone app, ~100 staff, full document intake. Single-AZ, with backups and snapshots doing the recovery work.

Line item	Specification	Calculation	USD/mo	₹/mo
Amazon RDS for PostgreSQL	db.t4g.small, Single-AZ	$\$0.042 \times 730 \text{ hr}$	30.66	2,897
RDS storage	100 GiB gp3	$100 \times \$0.131$	13.10	1,238
RDS backups beyond free	50 GiB extra retention	$50 \times \$0.095$	4.75	449
EC2 application server	t4g.medium, Linux	$\$0.0224 \times 730 \text{ hr}$	16.35	1,545
EBS volume	60 GiB gp3	$60 \times \$0.0912$	5.47	517
EBS snapshots	~40 GiB retained	$40 \times \$0.05$	2.00	189
S3 (invoice scans)	200 GB Standard	$200 \times \$0.025$	5.00	472
S3 requests	moderate volume	estimate	0.75	71
Public IPv4 address	1 address	$\$0.005 \times 730 \text{ hr}$	3.65	345
Total, on-demand, before tax			\$95.16	₹8,993
Including 18% GST			—	₹10,612

Line item	Specification	Calculation	USD/mo	₹/mo
Data transfer out	200 GB, first 100 free	$100 \times \$0.1093$	10.93	1,033
CloudWatch logs and alarms	basic monitoring	estimate	2.50	236
Total, on-demand, before tax			\$95.16	₹8,993
Including 18% GST			—	₹10,612

THIS IS THE ₹10K BUDGET, MET

And with a 1-year reserved instance on the two compute lines it comes down further — see Section 5. Commit after the first month, once the real CPU numbers are in.

4 Plan C — High availability NO SINGLE POINT OF FAILURE

Multi-AZ database with an automatic standby in a second Availability Zone, larger application server, longer retention. For when a day of downtime costs more than the difference.

Line item	Specification	Calculation	USD/mo	₹/mo
Amazon RDS for PostgreSQL	db.t4g.medium, Multi-AZ	$\$0.167 \times 730 \text{ hr}$	121.91	11,520
RDS storage	200 GiB gp3, Multi-AZ rate	$200 \times \sim \$0.262$	52.40	4,952
RDS backups beyond free	100 GiB extra retention	$100 \times \$0.095$	9.50	898
EC2 application server	t4g.large, Linux	$\$0.0448 \times 730 \text{ hr}$	32.70	3,091
EBS volume	100 GiB gp3	$100 \times \$0.0912$	9.12	862
EBS snapshots	~60 GiB retained	$60 \times \$0.05$	3.00	284
S3 (invoice scans)	500 GB Standard	$500 \times \$0.025$	12.50	1,181
S3 requests	higher volume	estimate	1.50	142
Public IPv4 address	1 address	$\$0.005 \times 730 \text{ hr}$	3.65	345
Data transfer out	500 GB, first 100 free	$400 \times \$0.1093$	43.72	4,132
CloudWatch logs and alarms	detailed monitoring	estimate	5.00	472
Total, on-demand, before tax			\$295.00	₹27,878
Including 18% GST			—	₹32,896

If you also want the application server itself made redundant, add a second EC2 instance and an Application Load Balancer. Price that pair from the AWS calculator when you decide to do it — a load balancer changes the deployment shape (sticky sessions for the Flask POS login), not just the bill.

SIDE BY SIDE

Plan	Database	App server	USD/mo	₹/mo	₹/mo incl. GST
A — Pilot	db.t4g.micro, 20 GiB	t4g.small	35.06	3,313	3,910
B — Production	db.t4g.small, 100 GiB	t4g.medium	95.16	8,993	10,612
C — High availability	db.t4g.medium Multi-AZ, 200 GiB	t4g.large	295.00	27,878	32,896

Converted at ₹94.5 to the US dollar (mid-market, 4 September 2026). AWS India invoices in INR at its own rate, so expect a small difference either way; budget at ₹95-97.

5 Bringing the cost down

RESERVED INSTANCES AND SAVINGS PLANS

Compute is the biggest line in every plan, and AWS discounts it heavily in exchange for a commitment. Reserved Instances apply to RDS; Savings Plans and RIs both apply to EC2. AWS publishes savings of up to 69% for a three-year all-upfront reservation against on-demand. Indicatively, for Plan B's two compute lines (\$47.01/month on demand):

Commitment	Typical discount	Compute USD/mo	Plan B total USD/mo	₹/mo incl. GST
On-demand (no commitment)	—	47.01	95.16	10,612
1-year, no upfront	~33%	~31.50	~79.65	~8,882
3-year, all upfront	~55%	~21.15	~69.30	~7,728

READ THIS BEFORE COMMITTING

The discount percentages above are indicative. The console shows the exact reserved price for your chosen class, term and payment option *before* you confirm the purchase — use that number, not this table, for the actual decision. A reservation is a billing commitment for the full term whether or not the instance runs, so buy it after a month of real usage, not on day one. RDS reservations do carry size flexibility within an instance family, so a t4g reservation still applies if you later resize within t4g.

OTHER LEVERS, IN ORDER OF HOW MUCH THEY RETURN

- **Put S3 behind CloudFront** if invoice images are served to browsers often. Data transfer from S3 to CloudFront is free, and CloudFront's egress rate is lower than EC2's.
- **Right-size the storage, not just the instance.** 100 GiB of gp3 is ₹1,238/month whether it holds 8 GB or 98 GB — but it also buys you 100 GiB of free backup allowance, so do not shrink it below your backup footprint.
- **Move invoice images out of the database volume into S3.** S3 at \$0.025/GB is roughly a fifth of gp3 at \$0.131/GB, and it removes the growth pressure on the RDS disk.
- **Stop non-production instances overnight.** A dev database run 10 hours a day on weekdays costs about 30% of a full month.
- **Lifecycle old snapshots.** Snapshots you never delete are a bill that only grows.

6 The alternative: Aurora Serverless v2

Aurora PostgreSQL Serverless v2 scales capacity by the second and can scale to zero. It is worth knowing about, and worth ruling out deliberately. In Mumbai it is \$0.18 per ACU-hour (Aurora Standard), \$0.11 per GB-month of storage, and \$0.22 per million I/O requests. One ACU is roughly 2 GiB of memory.

Scenario	Compute	USD/mo	Comparable RDS	USD/mo
Averaging 1 ACU (~2 GiB)	1 × \$0.18 × 730	131.40	db.t4g.small	30.66
Averaging 2 ACU (~4 GiB)	2 × \$0.18 × 730	262.80	db.t4g.medium	61.32

Verdict: stay on RDS. Aurora Serverless wins when a database is idle most of the day and spikes hard occasionally. Essa's is the opposite — a shop and warehouse open twelve hours a day, six days a week, with steady traffic throughout. For that duty cycle Aurora costs roughly four times a fixed

t4g instance for the same memory, and adds per-I/O charges the RDS gp3 model does not have. Revisit only if the usage pattern turns genuinely intermittent.

7 Cost traps to watch for

Trap	What happens
T4g CPU credits	T-class instances run in Unlimited mode. Above the baseline CPU, sustained load is billed at \$0.075 per vCPU-hour — the same for every T size, in every region, and <i>not</i> covered by a Reserved Instance. A 2-vCPU instance pinned at high CPU for a full month would add up to ~\$109. Put a CloudWatch alarm on <code>CPUCreditBalance</code> and you will never be surprised by it.
Data transfer out	Usually the line nobody forecast. 100 GB per month is free across the whole account; after that it is \$0.1093/GB from Mumbai. Serving invoice images and label sheets to browsers is what drives it.
Backups past 100%	Free backup storage equals your provisioned storage. Push retention to 30 days on a busy database and you cross it. \$0.095/GiB-month after.
Snapshots after deletion	Delete the instance and the snapshots keep billing at \$0.095/GiB-month until you delete them too.
Public IPv4	Every public IPv4 address is \$0.005/hr — about ₹345 a month each, including ones attached to instances you have stopped.
NAT Gateway	If you place the app in a private subnet and it needs outbound internet (the vision API does), a NAT Gateway costs roughly \$0.056/hr plus per-GB processing — around ₹4,000/month before traffic. A public subnet with a tight security group avoids it for a single-server deployment.
RDS Extended Support	Stay on a PostgreSQL major version past its community end-of-life and AWS bills per vCPU-hour to keep patching it. Plan a major-version upgrade every couple of years instead.
Idle non-production	A forgotten staging database is a full monthly charge for zero use. RDS also auto-restarts a stopped instance after seven days.

8 Free tier, tax and billing

FREE TIER

AWS has two free-tier models running side by side, and which one applies depends on when the account was created. Older accounts get twelve months of 750 hours per month of a Single-AZ `db.t3.micro` / `db.t4g.micro`, 20 GB of gp2 storage and 20 GB of backup. Newer accounts get sign-up credits instead — \$100 at sign-up and up to \$100 more as you complete activities, usable across eligible services including RDS. **Check which one your account has before sizing around it**, and either way do not build production on free-tier hardware: a `db.t4g.micro` is a pilot instance, and the tier expires.

GST AND INR BILLING

- An Indian entity is billed by AISPL in INR, with **18% GST** on top of the figures in every table above.
- Add Essa's GSTIN (`33AADCE6591N1Z7`) to *Billing* → *Tax settings* before the first invoice so the GST is claimable as input credit. Adding it later does not retrospectively fix earlier invoices.
- AWS converts at its own rate on the invoice date, so the rupee figures here will move a little month to month. Budget at ₹95–97 to the dollar.

9 Recommendation

1. **Start on Plan B** — `db.t4g.small` Single-AZ with 100 GiB gp3, and a `t4g.medium` application server. About ₹10,600 a month including GST, which is the budget, and it runs the full business rather than a pilot.
2. **Set a budget alarm at ₹12,000** on day one, with alerts at 80% and 100%.
3. **Run on-demand for the first month.** Then look at CloudWatch: CPU utilisation, credit balance, connection count and actual data transfer out. That is the evidence for both the reservation decision and the sizing one.
4. **Buy a 1-year reservation in month two** if the numbers are steady — around ₹8,900/month including GST, at the exact rate the console quotes you.
5. **Move to Plan C when downtime starts costing real money**, not before. Multi-AZ doubles the database bill to buy automatic failover; a Single-AZ instance with 30-day backups and pre-deployment snapshots is an honest posture for a business this size, provided the restore has been rehearsed at least once.

Prepared by Phoenixx IT for Essa Garments Private Limited. All rates read from AWS's published pricing pages for Asia Pacific (Mumbai) on 4 September 2026 and are exclusive of taxes unless stated. Monthly figures use 730 hours. AWS changes prices without notice — reconfirm in the AWS Pricing Calculator before commercial commitment. Companion document: *Essa ERP — AWS Database Connection Guide*.